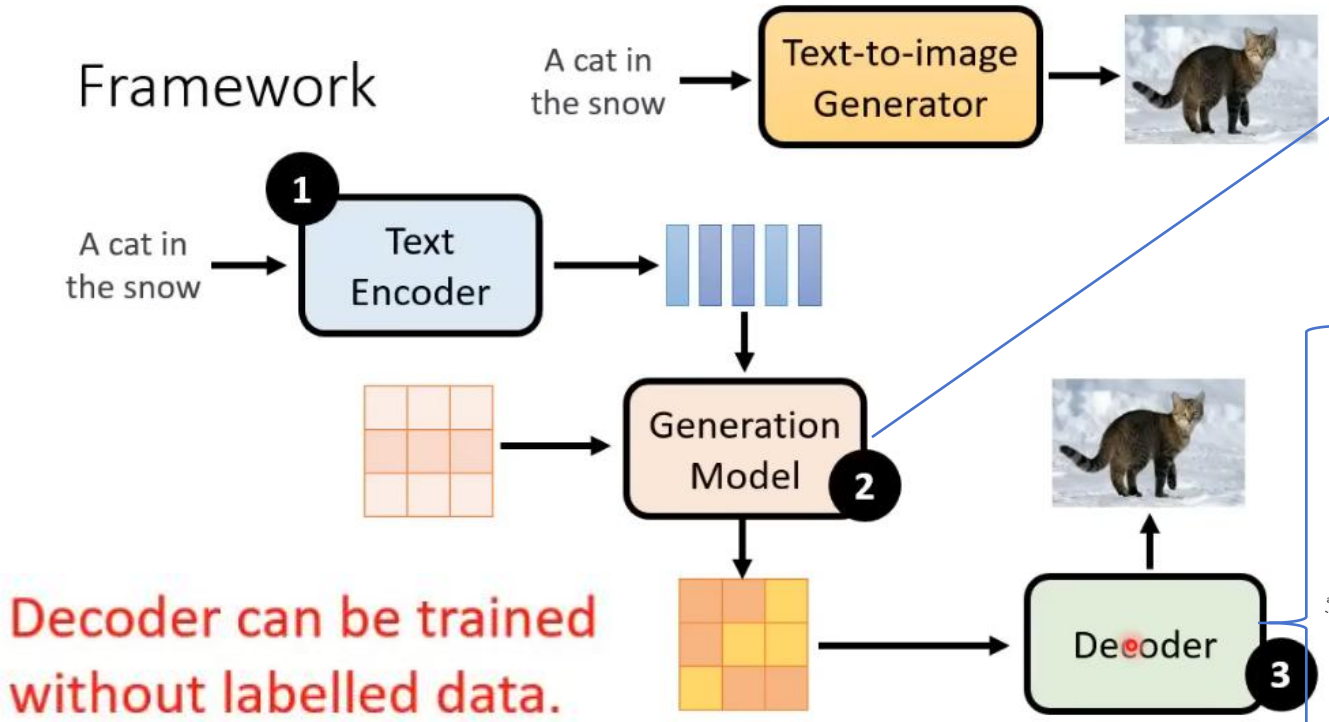
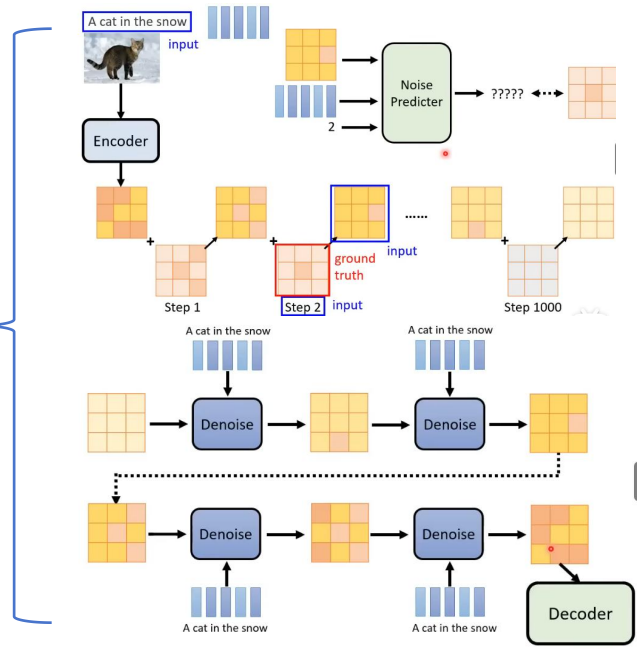
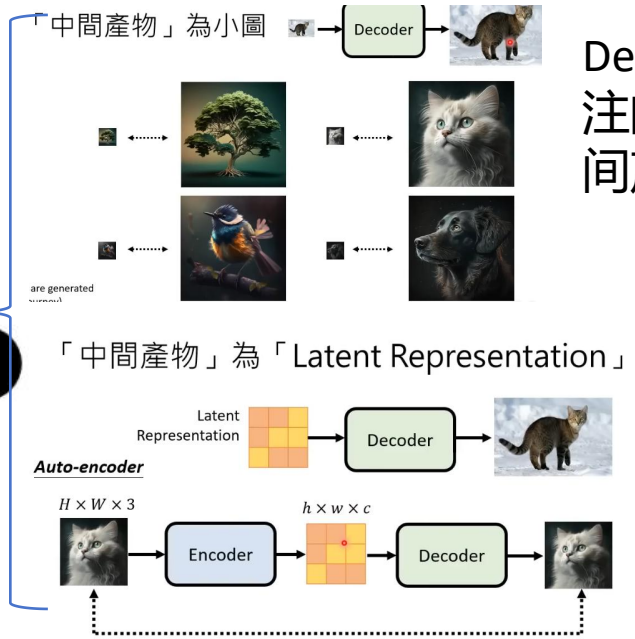
Diffusion Model

26.3.4

Stable diffusion

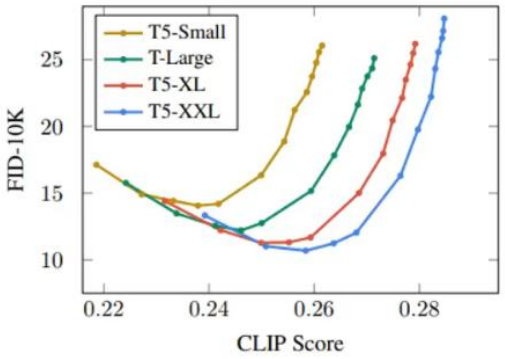




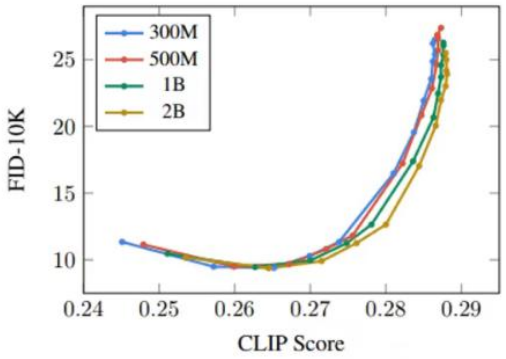


Decoder 可以没有用没有标注的数据训练，自己生成中间产物

evaluating the results of diffusion model



(a) Impact of encoder size.

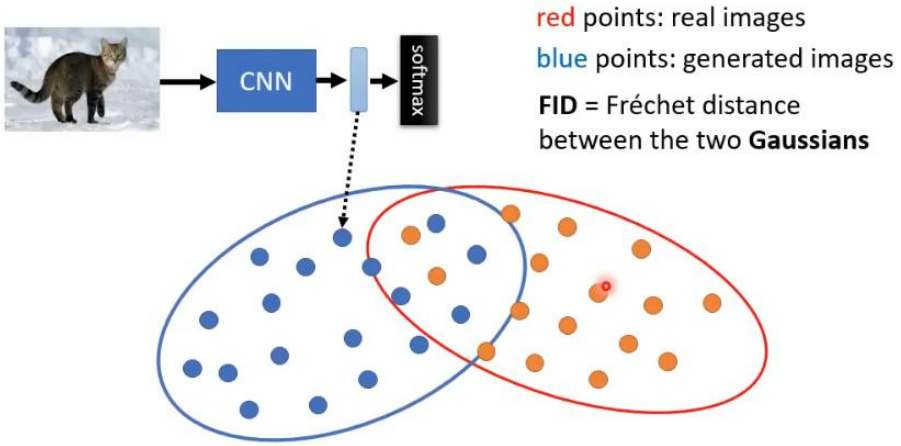


(b) Impact of U-Net size.

https://arxiv.org/abs/2205.11487

Fréchet Inception Distance (FID)

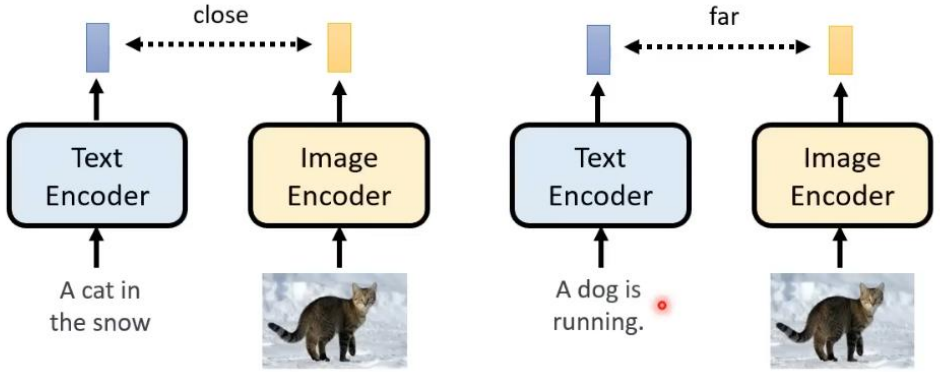
https://arxiv.org/abs/1706.08500



Contrastive Language-Image Pre-Training (CLIP)

https://arxiv.org/abs/2103.00020

400 million image-text pairs



Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t) \right\|^2$$
6: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

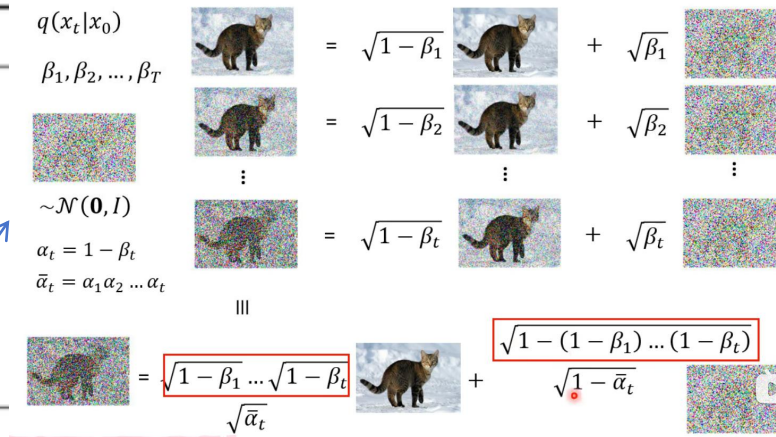
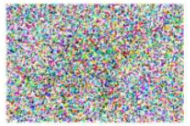


Figure 4: The training and sampling algorithms in DDPM (Image source: Ho et al. 2020)

x_0: clean image



ε: noise

Algorithm 1 Training

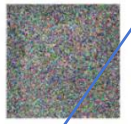
1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ ⟵ sample clean image
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ⟵ sample a noise
5: Take gradient descent step on
$$\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\underbrace{\boxed{\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}}}_{\text{Noisy image}}, t) \right\|^2$$
6: **until** converged

Target Noise

Noise predictor

$\bar{\alpha}_1, \bar{\alpha}_2, \ldots \bar{\alpha}_T$ → smaller

Inference



x_T

Algorithm 2 Sampling

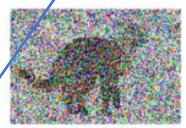
1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $\boxed{\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})}$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

sample a noise?!

DDPM

$\bar{\alpha}_1, \bar{\alpha}_2, \ldots \bar{\alpha}_T$

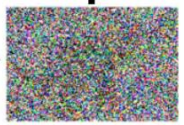
$\alpha_1, \alpha_2, \ldots \alpha_T$



x_t

t

Noise Predicter



$\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}$

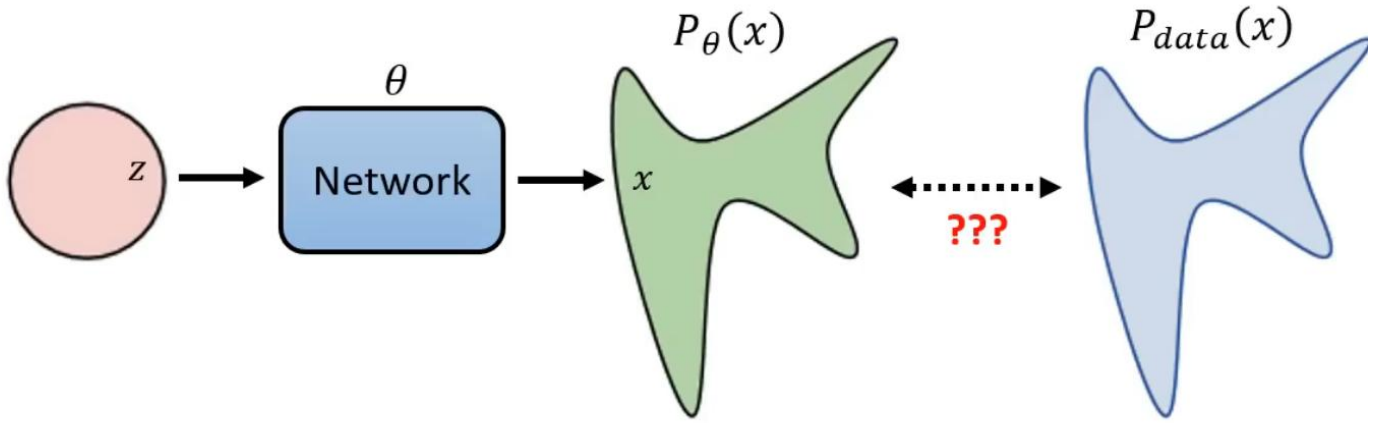
$\frac{1}{\sqrt{\alpha_t}}$

x_{t-1}

z

train的时候一步到位，而不是一步一步加噪

Maximum Likelihood Estimation



Sample $\{x^1, x^2, \dots, x^m\}$ from $P_{data}(x)$

We can compute $P_\theta(x^i)$

???

$$\theta^* = arg\max_\theta \prod_{i=1}^{m} P_\theta(x^i)$$

Sample $\{x^1, x^2, \dots, x^m\}$ from $P_{data}(x)$

$$\theta^* = arg\max_\theta \prod_{i=1}^{m} P_\theta(x^i) \quad = arg\max_\theta log \prod_{i=1}^{m} P_\theta(x^i)$$

$$= arg\max_\theta \sum_{i=1}^{m} logP_\theta(x^i) \approx arg\max_\theta E_{x\sim P_{data}}[logP_\theta(x)]$$

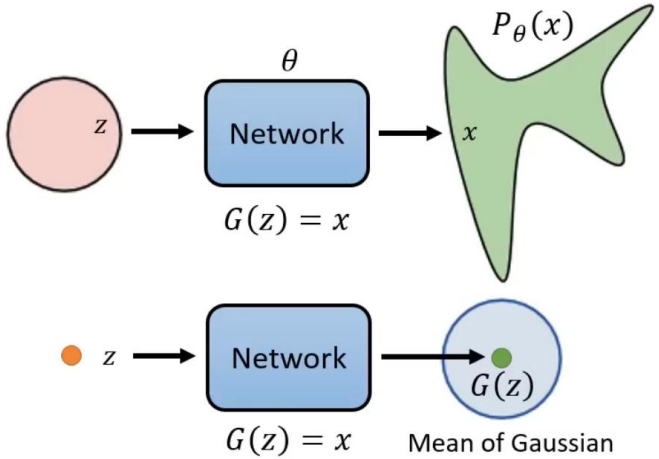
$$= arg\max_\theta \int_x P_{data}(x)logP_\theta(x)dx \quad - \int_x P_{data}(x)logP_{data}(x)dx$$

(not related to θ

$$= arg\max_\theta \int_x P_{data}(x)log\frac{P_\theta(x)}{P_{data}(x)}dx \quad = arg\min_\theta KL(P_{data}||P_\theta)$$

VAE

VAE: Compute $P_\theta(x)$



$$P_\theta(x) = \int_z P(z)P_\theta(x|z)dz$$

$$P_\theta(x|z) = \begin{cases} 1, & G(z) = x \\ 0, & G(z) \neq x \end{cases}$$

可能會幾乎都是 0 ☹

$$P_\theta(x|z) \propto exp(-\|G(z) - x\|_2)$$

VAE: Lower bound of $logP(x)$

$$logP_\theta(x) = \int_z q(z|x)logP(x)dz$$

$q(z|x)$ can be any distribution

$$= \int_z q(z|x)log\left(\frac{P(z,x)}{P(z|x)}\right)dz = \int_z q(z|x)log\left(\frac{P(z,x)}{q(z|x)}\frac{q(z|x)}{P(z|x)}\right)dz$$

$$= \int_z q(z|x)log\left(\frac{P(z,x)}{q(z|x)}\right)dz + \underbrace{\int_z q(z|x)log\left(\frac{q(z|x)}{P(z|x)}\right)dz}_{KL(q(z|x)||P(z|x))}$$

≥ 0

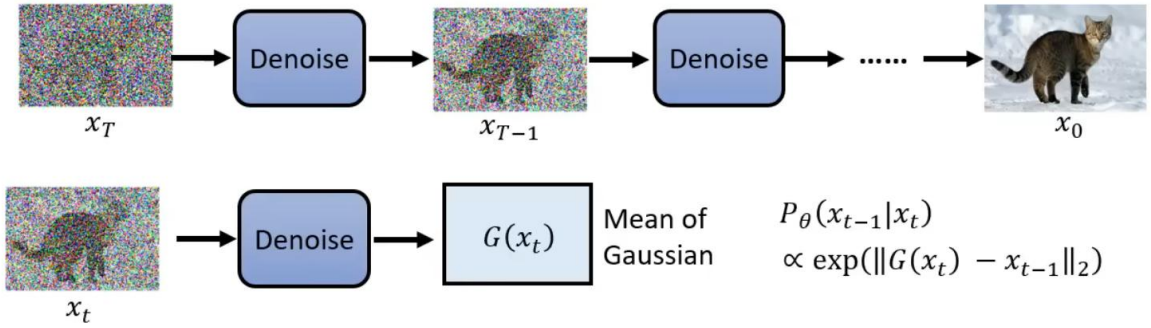
$$\geq \int_z q(z|x)log\left(\frac{P(z,x)}{q(z|x)}\right)dz = \mathrm{E}_{q(z|x)}\left[log\left(\frac{P(x,z)}{q(z|x)}\right)\right]$$

Encoder

lower bound

DDPM

DDPM: Compute $P_\theta(x)$



Mean of Gaussian

$$P_\theta(x_{t-1}|x_t) \propto \exp(\|G(x_t) - x_{t-1}\|_2)$$

$$P_\theta(x_0) = \int_{x_1:x_T} P(x_T)P_\theta(x_{T-1}|x_T)\dots P_\theta(x_{t-1}|x_t)\dots P_\theta(x_0|x_1)dx_1:x_T$$

$$\mathrm{E}_{q(x_1|x_0)}[logP(x_0|x_1)] - KL(q(x_T|x_0)||P(x_T))$$

$$\boxed{-\sum_{t=2}^{T}\mathrm{E}_{q(x_t|x_0)}[KL(q(x_{t-1}|x_t,x_0)||P(x_{t-1}|x_t))]}$$

minimize this

都是Gaussian

DDPM: Lower bound of $logP(x)$

VAE Maximize $logP_\theta(x)$ → Maximize $\mathrm{E}_{q(z|x)}[log\left(\frac{P(x,z)}{q(z|x)}\right)]$

Encoder

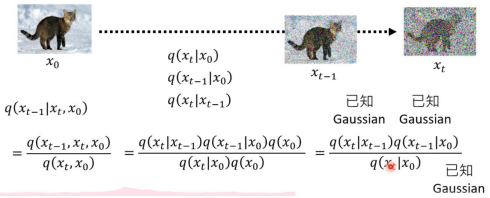
DDPM Maximize $logP_\theta(x_0)$ → Maximize $\mathrm{E}_{q(x_1:x_T|x_0)}[log\left(\frac{P(x_0:x_T)}{q(x_1:x_T|x_0)}\right)]$

Forward Process
(Diffusion Process)

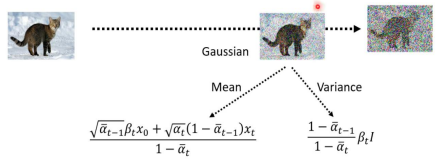
$$q(x_1:x_T|x_0) = q(x_1|x_0)q(x_2|x_1)\dots q(x_T|x_{T-1})$$

$$\mathrm{E}_{q(x_1|x_0)}[logP(x_0|x_1)] - KL(q(x_T|x_0)||P(x_T))$$

$$-\sum_{t=2}^{T}\mathrm{E}_{q(x_t|x_0)}[KL(q(x_{t-1}|x_t,x_0)||P(x_{t-1}|x_t))]$$



$$q(x_{t-1}|x_t,x_0) = \frac{q(x_{t-1},x_t,x_0)}{q(x_t,x_0)} = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)q(x_0)}{q(x_t|x_0)q(x_0)} = \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)}$$

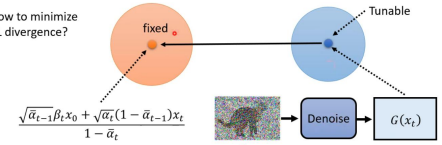


$$\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t x_0 + \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t}{1-\bar{\alpha}_t} \qquad \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t I$$

$$\mathrm{E}_{q(x_1|x_0)}[logP(x_0|x_1)] - KL(q(x_T|x_0)||P(x_T))$$

$$-\sum_{t=2}^{T}\mathrm{E}_{q(x_t|x_0)}[KL(q(x_{t-1}|x_t,x_0)||P(x_{t-1}|x_t))]$$

How to minimize KL divergence?



$$\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t x_0 + \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t}{1-\bar{\alpha}_t}$$

要让两个分布尽可能相近

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\varepsilon$$

$$x_t - \sqrt{1-\bar{\alpha}_t}\varepsilon = \sqrt{\bar{\alpha}_t}x_0$$

$$\frac{x_t - \sqrt{1-\bar{\alpha}_t}\varepsilon}{\sqrt{\bar{\alpha}_t}} = x_0$$

$$\frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t\frac{x_t - \sqrt{1-\bar{\alpha}_t}\varepsilon}{\sqrt{\bar{\alpha}_t}} + \sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t}{1-\bar{\alpha}_t}$$

$$= \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon\right)$$

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t\mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

只有ε是需要训练的，而αβ都是提前设定的常数